Coverage-Calibrated Candidate Reduction for Repeated Mixed-Integer Optimization: Finite-Sample Exactness, Smooth Recovery, and Certified Optimization

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Abstract

Repeated mixed-integer optimization (MIO) problems arising in rolling-horizon control, scheduling, and multi-period portfolio rebalancing require solving sequences of NP-hard programs where combinatorial branch-and-bound exploration across high-dimensional decision spaces forms the primary computational bottleneck. We propose a rigorous mathematical framework of *coverage-calibrated candidate reduction* that systematically exploits the temporal regularity of active combinatorial structures. By physically projecting the canonical N-dimensional MIO onto a B-dimensional nested candidate subproblem ($B \ll N$), solvers are invoked over dramatically reduced domains while retaining full mathematical authority over operational constraints. We establish the foundational concept of the candidate-family-dependent *Exactness Budget* $B^{\dagger}_{\mathcal{C}}(\theta)$—the minimal candidate dimension guaranteeing exact objective recovery—and prove an exactness-threshold monotonicity theorem alongside an exact support-entry characterization invariant to multiple global optima. Leveraging split-conformal prediction over repeated instance streams, we derive distribution-free and instance-adaptive finite-sample candidate-budget rules guaranteeing that the restricted problem recovers the exact full-space optimum with probability at least $1 - \alpha$ under exchangeability. For instances where initial predictions err, we prove that geometric candidate expansion ladders require at most logarithmic recovery stages, bounding cumulative structural work to $O(B^{\dagger})$. A deterministic fail-safe layer, strengthened via perspective-formulation continuous lower bounds, provides a posteriori solver-conditioned certification or triggers systematic ladder expansion. Evaluations across integer-parametric facility location benchmarks and multi-period portfolio MIQPs delineate scale-dependent computational gains alongside clear operational boundaries: while candidate reduction delivers an aggregate $1.18\times$ end-to-end throughput speedup at $N = 50$, recovery overhead at intermediate scales ($N = 30$) induces an end-to-end slowdown ($0.85\times$), certified 90–95% coverage collapses to the full universe in small instances, and perspective relaxation tightening fails to achieve wall-clock break-even at tested scales due to continuous conic solve overhead. Finally, an integer-lattice exact-oracle study exhibits empirical coverage (89.00% at $\alpha = 0.20$, 95% bootstrap CI $[83.33\%, 94.00\%]$) consistent with finite-sample theory, and a 2024 rolling-horizon MPC audit confirms practical feasibility under empirical temporal dependence. **Keywords:**

Repeated mixed-integer optimization, Conformal prediction, Learning-augmented optimization, Candidate reduction, Perspective relaxation, Portfolio optimization, Model predictive control.

1 Introduction

Mixed-integer optimization (MIO) provides an expressive paradigm for modeling complex operational systems subject to coupled continuous dynamics and discrete logical constraints. In receding-horizon model predictive control (MPC), sequential scheduling, and quantitative finance, MIO instances must be solved repeatedly as new state observations arrive. For instance, multi-period portfolio rebalancing subject to fixed transaction fees, minimum charged trade floors, discrete cardinality limits, and liquidity bounds yields large-scale mixed-integer quadratic programs (MIQPs). As the nominal entity universe N and forward planning horizon H expand, the discrete combinatorial search space grows as $2^{O(NH)}$, rendering repeated branch-and-bound exploration computationally prohibitive in latency-sensitive environments.

In repeated mathematical programming, successive problem instances θ_t are not independent draws from unconstrained distributions; rather, underlying physical states, transition dynamics, and covariance structures induce substantial regularity in the active combinatorial active sets. While recent learning-to-optimize paradigms have explored neural heuristics to predict continuous allocations or end-to-end discrete decisions, such methods sacrifice feasibility, violate hard operational bounds, and forfeit mathematical optimality certificates. Conversely, internal solver learning (e.g., learned branching or cutting plane selection) retains monolithic model dimensions, leaving root relaxation solving and large tableau updates unmitigated.

This paper establishes an alternative, mathematically rigorous framework: *coverage-calibrated candidate reduction*. Rather than replacing the mathematical optimizer, we use machine-learned signals or convex relaxations strictly to rank discrete entities, physically projecting the canonical N-dimensional problem onto a tractable B-dimensional candidate subproblem ($B \ll N$). Within this candidate subproblem, an off-the-shelf mathematical solver is retained with full authority, strictly enforcing all operational constraints, while mandatory inclusion of current operational states preserves state feasibility and liquidation paths.

1.1 The Central Theoretical Challenge and Core Message

The fundamental theoretical challenge in candidate reduction is *candidate budget calibration*: if budget B is chosen too small, the reduced problem omits optimal combinatorial supports, leading to objective degradation; if B is set too large, computational acceleration vanishes. Hitherto, candidate reduction has relied on heuristic budget tuning without statistical or mathematical guarantees.

This paper resolves this trade-off by establishing a coherent theoretical spine:

> *Statistical learning and finite-sample conformal calibration are harnessed to determine the minimal candidate budget required for exact objective recovery with guaranteed probability $1 - \alpha$, while deterministic continuous-relaxation certificates and geometric recovery ladders protect unconditionally against prediction failure.*

1.2 Contributions

The primary contributions of this paper are structured from foundational optimization theory to cross-domain computational characterization:

C1. Generic Exactness-Budget and Support-Entry Theory (Sections 3–4):
We formulate repeated MIO under nested candidate families and introduce the *Exactness Budget* $B^{\dagger}_{\mathcal{C}}(\theta)$—the minimal candidate dimension achieving exact full-problem objective recovery. We prove the Exactness-Threshold Monotonicity Theorem, showing that $B \geq B^{\dagger}_{\mathcal{C}}(\theta)$ is both necessary and sufficient for objective exactness, and prove a sharp upper bound via optimal support depth $R^{\dagger}(\theta)$ that remains rigorous under multiple global optima.

C2. Finite-Sample Candidate-Dimension Calibration (Sections 4.3–4.4):
Under explicit instance exchangeability, we apply split-conformal calibration to derive both marginal and instance-adaptive closed-form candidate-budget rules. We prove that the resulting restricted problem achieves exact objective recovery with non-asymptotic probability at least $1 - \alpha$, establishing a principled bridge between statistical learning theory and mixed-integer programming without asymptotic approximations.

C3. Exactness-Reaching Recovery and Structural Work Bounds (Section 5):
For instances where initial candidate budgets fall short, we analyze a geometric ladder recovery schedule and prove that the required recovery depth grows logarithmically with candidate-budget error ($O(\log_r(B^{\dagger}/B_1))$). We further prove that cumulative instantiated candidate dimensions, variables, and constraints across the entire recovery trajectory are strictly bounded by $O(B^{\dagger})$, establishing structural model-size complexity bounds without heuristic runtime claims.

C4. Solver-Conditioned Certification and Perspective Dominance (Section 5.4):
We integrate a deterministic fail-safe layer combining continuous relaxation lower bounds and full-universe fallback. For quadratic objectives, we derive a valid perspective relaxation decomposition ($\boldsymbol{\Sigma} = \boldsymbol{D} + \boldsymbol{R}$) and prove lower-bound dominance ($\mathrm{LB}^{\mathrm{basic}} \leq \mathrm{LB}^{\mathrm{persp}} \leq J^{*}$). Importantly, we treat computational efficacy as an empirical question, demonstrating that bound tightening alone does not guarantee wall-clock acceleration.

C5. Cross-Domain Empirical Frontier and Operational Characterization (Sections 6–9):
We instantiate the framework across both multi-period portfolio MIQPs and scalable integer-parametric facility location benchmarks. We establish an integer-lattice exact-oracle benchmark that validates finite-sample exactness theory, evaluate the confidence–compression–speedup frontier, and demonstrate through a 2024 rolling-horizon backtest on US equities how the dual-layer architecture maintains optimization authority under empirical market dynamics.

1.3 Scope and Non-Claims

We explicitly do not claim investment-performance superiority over passive benchmarks, nor do we assert that supervised learning outperforms continuous relaxations at support prediction

in all regimes. Furthermore, we make no empirical optimality claims for large nominal universes ($N \geq 100$) where restricted solver licenses prevent unreduced baseline solves. Our focus is strictly computational and methodological: characterizing the speed, node reduction, and solution fidelity of candidate-reduced mixed-integer optimization.

2 Related Work and Novelty Boundary

This work interfaces mathematical programming, model predictive control, and machine learning for combinatorial optimization. We review relevant literature across these areas, present a direct comparison against the closest related works, and synthesize the positioning in Table 1.

2.1 Learning-Augmented Optimization and ML for MIP

The framework of learning-augmented algorithms ("algorithms with predictions") integrates machine-learned advice into online and discrete algorithms to improve average-case efficiency while preserving worst-case robustness [Mitzenmacher and Vassilvitskii, 2022, Lykouris and Vassilvitskii, 2021]. In parametric MPC and repeated mathematical programming, strategy-prediction methods [Bertsimas and Stellato, 2021, 2022] predict active combinatorial bases to map instances directly to unconstrained linear systems, while Bertsimas and Kim [2023] use Optimal Policy Trees to prescribe discrete decisions. Concurrently, predict-then-optimize frameworks [Elmachtoub and Grigas, 2022, Mandi et al., 2020] embed optimization objectives into loss functions for continuous parameter prediction. In contrast, our architecture addresses the *combinatorial search space*: it predicts which decision variables belong to the optimal active support, keeping prediction modular and optimizer-directed.

A parallel literature uses machine learning to accelerate internal branch-and-bound operations in mixed-integer programming (MIP), including branching variable selection [Khalil et al., 2016, Gasse et al., 2019], node selection [He et al., 2014], cutting-plane selection [Tang et al., 2020], and primal heuristics [Chmiela et al., 2021, Nair et al., 2020]. While internal solver learning retains the monolithic problem dimension, our approach focuses on external dimension reduction: inactive variables and constraints are physically eliminated prior to solver instantiation, shrinking the root relaxation and search tree.

2.2 Variable Screening and Mixed-Integer Portfolio Optimization

In continuous convex optimization, safe feature screening provably eliminates inactive variables using dual bounds [El Ghaoui et al., 2012, Benidis et al., 2018]. Recently, Wada et al. [2026] derived analytical safe screening rules for single-period cardinality-constrained portfolio optimization via perspective relaxations and Fenchel duality. In non-convex mixed-integer settings, Triantafyllou and Papathanasiou [2024] developed learned dimensionality reduction for supply-chain MIPs by predicting binary assignments. For the richer multi-period fixed-cost formulation considered here, a priori lossless screening remains an open theoretical problem. Instead, our framework performs empirical candidate reduction and manages approximation risk ex post via valid continuous relaxation lower bounds and nested candidate recovery.

Table 1: Positioning relative to closest literature streams. Symbols: ✓ = fully addressed / guaranteed; ∘ = partially addressed / conditional; × = not addressed / not supported; N.A. = not applicable.

Literature Stream	Representative Works	Candidate Reduction	Nested Family	Support Learning	Minimal Budget $B^\dagger$	Finite-Sample Calibration	Objective Exactness	Deterministic Recovery	Perspective Strengthened
Scalable MIQP	Bertsimas and Cory-Wright [2022]	×	×	×	×	×	×	×	∘
Strategy-Based MPC	Bertsimas and Stellato [2021, 2022]	×	×	✓	×	×	×	✓	×
Prescriptive Trees	Bertsimas and Kim [2023]	×	×	✓	×	×	×	×	×
Learned MIP Dim. Red.	Triantafyllou and Papathanasiou [2024]	✓	×	✓	×	×	×	×	×
Safe Screening for MIQP	Wada et al. [2026]	✓	×	×	×	×	×	N.A.	✓
Conformal Optimization	Maragno et al. [2023], Laufer-Goldshtein et al. [2024]	×	×	×	×	✓	× (Feas.)	×	×
Algorithms with Pred.	Lykouris and Vassilvitskii [2021], Mitzenmacher and Vassilvitskii [2022]	∘	×	×	×	×	∘ (Comp.)	✓	×
This Work (Coverage-Calibrated)	**Proposed Framework**	✓	✓	✓	✓	✓	✓	✓	✓

2.3 Novelty Boundary and Literature Comparison

To the best of our knowledge, the closest existing approaches identified in our literature audit do not directly calibrate the minimum candidate dimension required for exact objective recovery. Specifically:

- **Not A Priori Screening**: Unlike safe screening methods [El Ghaoui et al., 2012, Wada et al., 2026] that compute dual bounds to discard provably inactive variables in continuous single-period models, our framework performs dimension reduction for non-convex multi-period fixed-cost MIQPs where a priori safe screening is open. We manage approximation error ex-post via conformal budget calibration and relaxation certificates.
- **Not Strategy Classification**: Unlike strategy-based MPC [Bertsimas and Stellato, 2021, 2022] which predicts active inequality combinations to reduce MIOs to unconstrained linear systems, our method retains a fully constrained, mathematically complete MIO over candidate assets. Operational constraints cannot be violated within the candidate universe.
- **Distinction from Conformal Optimization**: Existing conformal optimization literature [Maragno et al., 2023, Laufer-Goldshtein et al., 2024] calibrates constraint right-hand sides to ensure robust feasibility ($\mathbb{P}(x \in \mathcal{X}) \geq 1-\alpha$). In contrast, we calibrate the *combinatorial search budget* B to guarantee exact objective optimality ($\mathbb{P}(J(P(B)) = J^*) \geq 1-\alpha$).

3 Generic Repeated Mixed-Integer Optimization Framework

We formulate repeated mixed-integer optimization under an explicit entity-block variable architecture prior to portfolio specialization.

3.1 Entity-Block Variable Architecture and Formulation

Definition 3.1 (Entity-Block Variable Architecture)**.** Let an instance of a repeated mixed-integer optimization problem be parameterized by vector $\theta \in \Theta$. The complete decision variable vector is partitioned into $N+1$ blocks:

$$v = (v_0, v_1, \ldots, v_N), \tag{1}$$

where:

(i) $v_0 \in \mathbb{R}^{d_0} \times \mathbb{Z}^{m_0}$ denotes the *global variable block* containing shared system states (e.g., cash reserves, system-wide budget conservation, slack variables) that are never eliminated by candidate reduction;

(ii) $v_i = (x_i, z_i) \in \mathcal{V}_i := \mathbb{R}^{d_i} \times \mathcal{Z}_i$ denotes the *entity variable block* for discrete entity $i \in \mathcal{N} := \{1, \ldots, N\}$, where $x_i \in \mathbb{R}^{d_i}$ denotes continuous decisions and $z_i \in \mathcal{Z}_i \subseteq \mathbb{Z}^{m_i}$ denotes integer/binary indicators;

(iii) For each entity $i \in \mathcal{N}$, there exists a designated *null block* $v_i^0 = (x_i^0, z_i^0) \in \mathcal{V}_i$ representing the physical and logical exclusion of entity i from active operations (typically $x_i^0 = \mathbf{0}_{d_i}$ and $z_i^0 = \mathbf{0}_{m_i}$).

The canonical repeated optimization problem P_θ is defined as:

$$P_\theta : \quad \min_{v=(v_0, v_1, \ldots, v_N)} \ f_\theta(v) \quad \text{s.t.} \quad v \in \mathcal{F}_\theta \subseteq \mathcal{V}_0 \times \prod_{i=1}^{N} \mathcal{V}_i, \tag{2}$$

where f_θ is convex in continuous variables, and $\mathcal{F}_\theta$ denotes the coupled feasible domain. The optimal objective value of Problem (2) is denoted $J_\theta^* \in [-\infty, +\infty]$.

Definition 3.2 (Active Entity Support and Support Family)**.** For any feasible solution $v \in \mathcal{F}_\theta$, the *active entity support* is the set of entities whose variables deviate from their null block:

$$\mathcal{S}(v) := \{i \in \mathcal{N} : v_i \neq v_i^0\} \subseteq \mathcal{N}. \tag{3}$$

Let $\Omega_\theta^* := \arg\min_{v \in \mathcal{F}_\theta} f_\theta(v)$ denote the set of global optimal solutions. The *optimal support family* $\mathcal{S}_{\text{opt}}(\theta)$ is defined as the collection of active supports corresponding to all global optima:

$$\mathcal{S}_{\text{opt}}(\theta) := \{\mathcal{S}(v^*) \subseteq \mathcal{N} : v^* \in \Omega_\theta^*\}. \tag{4}$$

3.2 Nested Candidate Families and Subproblems

Definition 3.3 (Nested Candidate Family under Entity Blocks)**.** For instance $\theta \in \Theta$ and candidate family $\mathcal{C}_\theta$, let $B_{\min,\mathcal{C}}(\theta) \in \{0, 1, \ldots, N\}$ denote the minimal candidate size ensuring subproblem feasibility. The collection $\mathcal{C}_\theta : \{B_{\min,\mathcal{C}}(\theta), \ldots, N\} \to 2^{\mathcal{N}}$ is a *nested candidate family* if:

(i) **Cardinality**: $|\mathcal{C}_\theta(B)| = B$ for all $B \in \{B_{\min,\mathcal{C}}(\theta), \ldots, N\}$;

(ii) **Nested Monotonicity**: $B_{\min,\mathcal{C}}(\theta) \leq B_1 \leq B_2 \leq N \implies \mathcal{C}_\theta(B_1) \subseteq \mathcal{C}_\theta(B_2)$;

(iii) **Full Coverage**: $\mathcal{C}_\theta(N) = \mathcal{N}$;

(iv) **Mandatory State Preservation**: There exists a problem-specific mandatory entity set $\mathcal{M}_\theta \subseteq \mathcal{N}$ required for state feasibility such that $B_{\min,\mathcal{C}}(\theta) \geq |\mathcal{M}_\theta|$ and $\mathcal{M}_\theta \subseteq \mathcal{C}_\theta(B)$ for all $B \geq B_{\min,\mathcal{C}}(\theta)$.

For problems where the feasibility floor is invariant across rankings (such as portfolio rebalancing where baseline holdings fix $\mathcal{M}_\theta$), we write $B_{\min}(\theta)$ or $B_{\min}$ for conciseness.

Definition 3.4 (Candidate-Restricted Subproblem)**.** Given candidate set $\mathcal{C} \subseteq \mathcal{N}$, the physically reduced subproblem $P_\theta(\mathcal{C})$ restricts non-candidate entity blocks strictly to their designated null

values:

$$P_\theta(\mathcal{C}): \quad \min_{v=(v_0,v_1,\ldots,v_N)} f_\theta(v) \quad \text{s.t.} \quad v \in \mathcal{F}_\theta, \quad v_i = v_i^0 \ (\forall i \notin \mathcal{C}). \tag{5}$$

The optimal objective value is denoted $J_\theta(\mathcal{C})$ (defined as $+\infty$ if infeasible).

Assumption 3.5 (Baseline Solvability)**.** For every instance $\theta \in \Theta$: (a) $P_\theta(\mathcal{C}_\theta(B_{\min}))$ is feasible and f_θ is bounded from below on $\mathcal{F}_\theta$; (b) The global optimal solution set Ω_θ^* is non-empty, so that $\mathcal{S}_{\text{opt}}(\theta) \neq \emptyset$.

4 Exactness Budgets and Conformal Candidate Reduction

We now formalize the minimal candidate budget guaranteeing exact objective recovery and derive finite-sample coverage-calibrated budget rules.

4.1 Exactness Budget Definition

Definition 4.1 (Exactness Budgets and Operational Analogues)**.** Let $\mathcal{C}_\theta$ be an instance-dependent nested candidate family with feasibility floor $B_{\min,\mathcal{C}}(\theta)$.

(a) **Mathematical Exactness Budget** ($B_{\mathcal{C}}^\dagger(\theta)$): The minimal candidate budget achieving exact objective recovery:

$$B_{\mathcal{C}}^\dagger(\theta) \coloneqq \min\{B \in \{B_{\min,\mathcal{C}}(\theta), \ldots, N\} : J_\theta(\mathcal{C}_\theta(B)) = J_\theta^*\}. \tag{6}$$

(b) **Operational Reference-Fidelity Budget** ($B_{\mathcal{C},\varepsilon}^{\dagger,\text{ref}}(\theta)$): Given reference full-universe objective value J_θ^{ref} and non-negative tolerance $\varepsilon_{\text{ref}} \geq 0$:

$$B_{\mathcal{C},\varepsilon}^{\dagger,\text{ref}}(\theta) \coloneqq \min\{B \in \{B_{\min,\mathcal{C}}(\theta), \ldots, N\} : J_\theta(\mathcal{C}_\theta(B)) - J_\theta^{\text{ref}} \leq \varepsilon_{\text{ref}}\}. \tag{7}$$

(c) **Certified ε-Quality Budget** ($B_{\mathcal{C},\varepsilon}^{\dagger,\text{cert}}(\theta)$): Given a valid full-universe lower bound $\text{LB}_{\text{full}}^{\text{relax}}$ and non-negative tolerance $\varepsilon_{\text{cert}} \geq 0$:

$$B_{\mathcal{C},\varepsilon}^{\dagger,\text{cert}}(\theta) \coloneqq \min\{B \in \{B_{\min,\mathcal{C}}(\theta), \ldots, N\} : \text{UB}_\theta(\mathcal{C}_\theta(B)) - \text{LB}_{\text{full}}^{\text{relax}} \leq \varepsilon_{\text{cert}}\}. \tag{8}$$

Remark 4.2 (One-Sided Reference Fidelity and Strict Incumbent Improvement)*.* Definition 4.1(b) enforces a one-sided condition: the candidate-reduced objective must not exceed the reference incumbent by more than ε_{ref}. If the candidate subproblem identifies an incumbent strictly superior to a heuristic reference ($J_\theta(\mathcal{C}_\theta(B)) < J_\theta^{\text{ref}}$), this is automatically successful. Conversely, Definition 4.1(c) requires $\text{LB}_{\text{full}}^{\text{relax}}$ to be a mathematically valid lower bound for the *unrestricted full-universe* optimization problem, ensuring $0 \leq \text{UB} - J_\theta^* \leq \varepsilon_{\text{cert}}$.

Remark 4.3 (Relationship between Theory and Finite-Precision Solvers)*.* The Exactness Budget $B_{\mathcal{C}}^\dagger(\theta)$ is a mathematical oracle target. In numerical experiments, finite-precision MIP solvers produce reference-match and solver-certified ε-quality budgets because standard floating-point solves do not reveal exact real-valued objective equality. The former ($B^{\dagger,\text{ref}}$) evaluates empirical solution fidelity, whereas the latter ($B^{\dagger,\text{cert}}$) provides a rigorous ε-optimality guarantee ($0 \leq \text{UB} - J_\theta^* \leq \varepsilon_{\text{cert}}$) via a verified continuous relaxation lower bound.

Theorem 4.4 (Candidate Monotonicity and Exactness Threshold)**.** *Under Definition 3.3 and Assumption 3.5, for every $\theta \in \Theta$:*

(i) ***Objective Monotonicity****:*

$$B_{\min} \le B_1 \le B_2 \le N \implies J_\theta(\mathcal{C}_\theta(B_1)) \ge J_\theta(\mathcal{C}_\theta(B_2)) \ge J^*_\theta. \tag{9}$$

(ii) ***Threshold Exactness****: The budget $B^\dagger(\theta)$ exists uniquely in $\{B_{\min}, \dots, N\}$. For all $B \in \{B_{\min}, \dots, N\}$:*

$$B \ge B^\dagger(\theta) \iff J_\theta(\mathcal{C}_\theta(B)) = J^*_\theta. \tag{10}$$

(iii) ***Support-Containment Upper Bound****:*

$$B^\dagger(\theta) \le \min_{S^* \in \mathcal{S}_{\text{opt}}(\theta)} \min\{B \in \{B_{\min}, \dots, N\} : S^* \subseteq \mathcal{C}_\theta(B)\}. \tag{11}$$

Proof. See Appendix A.2. □

4.2 Optimal Support Depth

Let $\text{rank}_\theta(i) \in \{1, \dots, N - |\mathcal{M}_\theta|\}$ denote the ranking score of non-mandatory entity $i \in \mathcal{N} \setminus \mathcal{M}_\theta$ (1 being top rank).

Definition 4.5 (Optimal Support Depth)**.** The *optimal support depth* $R^\dagger(\theta)$ is the minimum ranking rank required to capture at least one globally optimal support:

$$R^\dagger(\theta) := \min_{S^* \in \mathcal{S}_{\text{opt}}(\theta)} \max_{i \in S^* \setminus \mathcal{M}_\theta} \text{rank}_\theta(i), \tag{12}$$

with $\max \emptyset := 0$ if there exists $S^* \in \mathcal{S}_{\text{opt}}(\theta)$ such that $S^* \setminus \mathcal{M}_\theta = \emptyset$.

Proposition 4.6 (Support-Depth Upper Bound on $B^\dagger$)**.** *When candidate sets are constructed by augmenting mandatory entities $\mathcal{M}_\theta$ with the top-ranked non-mandatory entities, the exactness budget satisfies:*

$$B^\dagger(\theta) \le |\mathcal{M}_\theta| + R^\dagger(\theta). \tag{13}$$

Proof. See Appendix A.3. □

4.3 Finite-Sample Marginal Calibrated Exactness

Assumption 4.7 (Instance Exchangeability)**.** The calibration instances $\theta_1, \dots, \theta_n$ and the new test instance θ_{n+1} are exchangeable random variables.

For calibration instances $j \in \{1, \dots, n\}$, let $B^\dagger_j := B^\dagger(\theta_j)$ denote their true exactness budgets. Let $B^\dagger_{(1)} \le B^\dagger_{(2)} \le \cdots \le B^\dagger_{(n)}$ denote the sorted order statistics.

Definition 4.8 (Finite-Sample Quantile Index and Marginal Budget Rule)**.** For user-specified miscoverage tolerance $\alpha \in (0, 1)$, define index $k_\alpha := \lceil (n+1)(1-\alpha) \rceil$. The *marginal calibrated*

candidate budget is defined as:

$$\widehat{B}_\alpha := \begin{cases} \min(N,\ B^\dagger_{(k_\alpha)}), & \text{if } k_\alpha \le n, \\ N, & \text{if } k_\alpha = n+1. \end{cases} \tag{14}$$

Theorem 4.9 (Finite-Sample Marginal Calibrated Exactness)**.** *Under Assumption 4.7, for any* $\alpha \in (0,1)$ *and calibration sample size* $n \ge 1$*:*

$$\mathbb{P}(B^\dagger(\theta_{n+1}) \le \widehat{B}_\alpha) \ge 1-\alpha, \tag{15}$$

and consequently, the candidate-restricted subproblem attains the exact full-problem optimal objective with guaranteed probability:

$$\mathbb{P}\Big(J_{\theta_{n+1}}(\mathcal{C}_{\theta_{n+1}}(\widehat{B}_\alpha)) = J^*_{\theta_{n+1}}\Big) \ge 1-\alpha. \tag{16}$$

Proof. See Appendix A.4. □

4.4 Instance-Adaptive Split-Conformal Exactness

Let $X_\theta \in \mathbb{R}^d$ denote an observable instance feature vector available prior to solving. We train a budget predictor $\hat{b} : \mathbb{R}^d \to \mathbb{R}$ on an independent training set $\mathcal{D}_{\text{train}}$ and calibrate on a disjoint set $\mathcal{D}_{\text{cal}} = \{(X_j, B^\dagger_j)\}_{j=1}^{n_{\text{cal}}}$. Algorithm 1 specifies the protocol.

Algorithm 1: Instance-Adaptive Split-Conformal Calibration Protocol

Input: Disjoint datasets $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{cal}} = \{(X_j, B^\dagger_j)\}_{j=1}^{n_{\text{cal}}}$; tolerance $\alpha \in (0,1)$; universe size N; minimum budget $B_{\min}$

Output: Deterministic adaptive candidate budget mapping $B_\alpha(\cdot)$

Fit budget prediction model $\hat{b} : \mathcal{X} \to \mathbb{R}$ exclusively on $\mathcal{D}_{\text{train}}$
On $\mathcal{D}_{\text{cal}}$, evaluate one-sided residuals: $R_j := B^\dagger_j - \hat{b}(X_j)$ for all $j \in \{1, \dots, n_{\text{cal}}\}$
Sort residuals in ascending order: $R_{(1)} \le R_{(2)} \le \cdots \le R_{(n_{\text{cal}})}$
Compute quantile index $k_{\text{cal}} \leftarrow \lceil (n_{\text{cal}}+1)(1-\alpha) \rceil$
if $k_{\text{cal}} \le n_{\text{cal}}$ **then**
    $q_\alpha \leftarrow R_{(k_{\text{cal}})}$
    Define $B_\alpha(X) := \min\Big(N,\ \max(B_{\min},\ \lceil \hat{b}(X) + q_\alpha \rceil)\Big)$
else
    Define safe fallback: $B_\alpha(X) := N$
return $B_\alpha(\cdot)$

Theorem 4.10 (Instance-Adaptive Split-Conformal Exactness)**.** *Suppose* $(X_j, B^\dagger_j)_{j \in \mathcal{D}_{\text{cal}}}$ *and* $(X_{\text{new}}, B^\dagger_{\text{new}})$ *are exchangeable conditional on* $\mathcal{D}_{\text{train}}$*. Then for any* $\alpha \in (0,1)$*, the adaptive candidate budget* $B_\alpha(X_{\text{new}})$ *returned by Algorithm 1 satisfies:*

$$\mathbb{P}\Big(J_{\text{new}}(\mathcal{C}_{\text{new}}(B_\alpha(X_{\text{new}}))) = J^*_{\text{new}} \,\Big|\, \mathcal{D}_{\text{train}}\Big) \ge \mathbb{P}(B^\dagger_{\text{new}} \le B_\alpha(X_{\text{new}}) \mid \mathcal{D}_{\text{train}}) \ge 1-\alpha. \tag{17}$$

Consequently, the marginal probability integrated over training data draws is also at least $1-\alpha$*.*

Proof. See Appendix A.5. □

5 Smooth Recovery, Structural Work Bounds, and Certified Optimization

When initial candidate budgets are insufficient, a nested recovery algorithm systematically expands the candidate universe. We establish logarithmic recovery bounds, cumulative structural work limits, and a two-layer guarantee.

5.1 Logarithmic Recovery Depth

Consider the geometric candidate schedule initialized at $B_1 \geq B_{\min}$ with expansion ratio $r > 1$:

$$B_\ell := \min(N,\ \lceil B_1 r^{\ell-1} \rceil), \qquad \ell = 1, 2, \ldots, L, \tag{18}$$

where $L := 1 + \lceil \log_r(N/B_1) \rceil$ guarantees $B_L = N$. Let $\ell^*(\theta) := \min\{\ell \in \{1, \ldots, L\} : B_\ell \geq B^\dagger(\theta)\}$ denote the first recovery stage attaining objective exactness.

Theorem 5.1 (Logarithmic Recovery-Depth Bound)**.** *For any instance $\theta \in \Theta$, the exactness-recovery stage index satisfies:*

$$\ell^*(\theta) \leq 1 + \left\lceil \log_r \left(\frac{B^\dagger(\theta)}{B_1} \right) \right\rceil_+, \tag{19}$$

where $[x]_+ := \max(0, x)$.

Proof. See Appendix A.6. □

5.2 Cumulative Structural Work Bounds

While NP-hard worst-case branch-and-bound runtimes cannot be bounded a priori, the physical size of mathematical models passed to the solver can be bounded strictly:

Theorem 5.2 (Cumulative Structural Work and Candidate Dimension Bound)**.** *Under geometric schedule (18), the cumulative candidate dimensions instantiated across recovery stages up to exactness satisfy:*

$$\sum_{\ell=1}^{\ell^*(\theta)} B_\ell < \left(\frac{r^2}{r-1} \right) B^\dagger(\theta) + \ell^*(\theta). \tag{20}$$

Consequently, if decision variables and constraints are affine in candidate dimension ($n_{\text{vars}}(B) = c_v B + k_v$, $n_{\text{constrs}}(B) = c_c B + k_c$), the total cumulative model size instantiated before exact recovery is strictly $O(B^\dagger(\theta))$.

Proof. See Appendix A.7. □

Remark 5.3 (Scope of Structural Bound). Theorem 5.2 is an explicit structural model-size bound; it does not claim a polynomial wall-clock runtime guarantee for mixed-integer programming.

5.3 Two-Layer Prediction-Robust Optimization Guarantee

Theorem 5.4 (Two-Layer Prediction-Robust Optimization Guarantee)**.** *Let an instance θ be solved by initializing Candidate-Restricted MIO at budget $B_1 = B_\alpha(X_\theta)$ (Algorithm 1) within the multi-stage certified recovery procedure. The returned incumbent $\hat{v}$ satisfies two decoupled layers of assurance:*

(I) ***Layer 1 (Probabilistic Fast-Path Exactness)****: Under exchangeability (Assumption 4.7), the initial candidate problem is exact with probability at least $1-\alpha$:*

$$\mathbb{P}\Big(J_\theta(\mathcal{C}_\theta(B_\alpha(X_\theta))) = J_\theta^*\Big) \geq 1-\alpha. \tag{21}$$

Executing the first-stage candidate solution immediately (Fast Mode) incurs zero objective degradation with probability at least $1-\alpha$.

(II) ***Layer 2 (Deterministic Quality Control)****: For any instance (including under predictor failure or distribution shift), the terminal solution $\hat{v}$ and upper bound $\mathrm{UB} = f_\theta(\hat{v})$ in Certified Mode satisfy:*

- *If termination occurs at stage $\ell \leq L$ with $\mathrm{UB} - \mathrm{LB}_{\mathrm{full}}^{\mathrm{relax}} \leq \varepsilon_{\mathrm{cert}}$, then $0 \leq f_\theta(\hat{v}) - J_\theta^* \leq \varepsilon_{\mathrm{cert}}$.*
- *If recovery reaches $B_L = N$ and the solver terminates with status `OPTIMAL`, then $0 \leq f_\theta(\hat{v}) - J_\theta^* \leq \varepsilon_{\mathrm{MIP}}|\mathrm{UB}|$ (or $\leq \varepsilon_{\mathrm{abs}}$ if $\mathrm{UB} = 0$).*
- *If the solve terminates without a proven bound certificate, $\hat{v}$ is explicitly designated as `UNCERTIFIED`. Reaching the full universe ($B = N$) does not in itself imply certification.*

Proof. See Appendix A.8. □

5.4 Perspective-Strengthened Lower Bounds

For quadratic objective terms $x^\top \boldsymbol{\Sigma} x$ with $\boldsymbol{\Sigma} \succeq 0$, continuous relaxations of semi-continuous constraints $x_i \leq U_i y_i$ with $y_i \in [0,1]$ leave quadratic costs independent of fractional binary indicators.

Definition 5.5 (Valid PSD Diagonal Extraction)**.** A matrix decomposition $\boldsymbol{\Sigma} = \boldsymbol{D} + \boldsymbol{R}$ is a *valid PSD diagonal extraction* if:

$$\boldsymbol{D} = \mathrm{diag}(d_1, \ldots, d_N) \succeq 0 \quad (\text{i.e. } d_i \geq 0, \forall i) \quad \text{and} \quad \boldsymbol{R} = \boldsymbol{\Sigma} - \boldsymbol{D} \succeq 0. \tag{22}$$

Proposition 5.6 (Deterministic Safe PSD Diagonal Construction)**.** *For any $\boldsymbol{\Sigma} \in \mathbb{S}_+^N$, let $\lambda_{\min}(\boldsymbol{\Sigma}) \geq 0$ denote its minimum eigenvalue. For parameter $\rho_{\mathrm{persp}} \in [0,1]$:*

$$\boldsymbol{D} \coloneqq \rho_{\mathrm{persp}} \lambda_{\min}(\boldsymbol{\Sigma}) \boldsymbol{I}_N, \qquad \boldsymbol{R} \coloneqq \boldsymbol{\Sigma} - \boldsymbol{D} \tag{23}$$

satisfies Definition 5.5 unconditionally.

Proof. $\boldsymbol{D} \succeq 0$ is immediate since $\rho_{\text{persp}} \geq 0$ and $\lambda_{\min} \geq 0$. For any unit vector u, $u^\top \boldsymbol{R} u = u^\top \boldsymbol{\Sigma} u - \rho_{\text{persp}} \lambda_{\min} u^\top \boldsymbol{I} u \geq \lambda_{\min} - \rho_{\text{persp}} \lambda_{\min} \geq 0$, proving $\boldsymbol{R} \succeq 0$. □

In the perspective formulation, we introduce auxiliary variables $t_i \geq 0$ subject to:

$$t_i y_i \geq x_i^2, \quad t_i \geq 0, \quad y_i \in [0, 1], \tag{24}$$

which is natively represented in mathematical solvers via rotated second-order cone constraints $2uv \geq w^2$ with $u = t_i, v = y_i/2, w = x_i$.

Theorem 5.7 (Perspective Lower-Bound Dominance)**.** *Let* LB^{basic} *denote the optimal value of the standard continuous relaxation, and let* LB^{persp} *denote the optimal value of the perspective-strengthened relaxation under Definition 5.5. Then:*

$$\text{LB}^{\text{basic}} \leq \text{LB}^{\text{persp}} \leq J^*. \tag{25}$$

Consequently, for any feasible upper bound UB, $\text{UB} - \text{LB}^{\text{persp}} \leq \text{UB} - \text{LB}^{\text{basic}}$, *and the set of instances certified by the strengthened relaxation weakly contains that of the basic relaxation:*

$$\{\theta : \text{UB}_\theta - \text{LB}_\theta^{\text{basic}} \leq \varepsilon_{\text{cert}}\} \subseteq \{\theta : \text{UB}_\theta - \text{LB}_\theta^{\text{persp}} \leq \varepsilon_{\text{cert}}\}. \tag{26}$$

Proof. See Appendix A.9. □

6 Multi-Period Portfolio MIQP Instantiation

We now specialize the generic framework to multi-period portfolio rebalancing under operational frictions.

6.1 Problem Formulation and Accounting

At decision epoch t, portfolio holdings and cash are expressed in decision-time numeraire wealth $V_t = 1.0$. Let $\mathcal{N} = \{1, \ldots, N\}$ index assets and $h \in \{0, \ldots, H-1\}$ index forward steps. The decision vector maps to the entity-block architecture: $v_0 = \{b_h^-, b_h, \kappa_h\}_{h=0}^{H-1}$ and $v_i = \{x_{h,i}^-, x_{h,i}, u_{h,i}, q_{h,i}, y_{h,i}, z_{h,i}\}_{h=0}^{H-1}$. The canonical MIQP is:

$$\min_{v} \quad \sum_{h=0}^{H-1} \beta^h \left[\frac{\gamma}{2} x_h^\top \mathbf{\Sigma}_h x_h - \mu_h^\top x_h - r_{f,h} b_h + \kappa_h \right] \tag{27a}$$

$$\text{s.t.} \quad x_0^- = \bar{w}_t, \qquad b_0^- = \bar{c}_t, \tag{27b}$$

$$x_{h,i} - u_{h,i} = x_{h,i}^-, \quad \forall h, i \tag{27c}$$

$$\mathbf{1}^\top x_h + b_h = \mathbf{1}^\top x_h^- + b_h^- - \kappa_h, \quad \forall h \tag{27d}$$

$$\kappa_h - \sum_{i=1}^{N} (a_{i,h} q_{h,i} + f_{i,h} z_{h,i}) = 0, \quad \forall h \tag{27e}$$

$$q_{h,i} - u_{h,i} \geq 0, \quad q_{h,i} + u_{h,i} \geq 0, \quad \forall h, i \tag{27f}$$

$$x_{h,i} \leq U_{i,h}\, y_{h,i}, \quad \forall h, i \tag{27g}$$

$$x_{h,i} \geq \ell_{i,h}\, y_{h,i}, \quad \forall h, i \tag{27h}$$

$$\sum_{i=1}^{N} y_{h,i} \leq K_h, \quad \forall h \tag{27i}$$

$$q_{h,i} \leq M_{i,h}^q\, z_{h,i}, \quad \forall h, i \tag{27j}$$

$$q_{h,i} \geq \delta_{i,h}\, z_{h,i}, \quad \forall h, i \tag{27k}$$

$$\sum_{i=1}^{N} q_{h,i} \leq \tau_h^{\max}, \quad \forall h \tag{27l}$$

$$x_{h+1,i}^- = g_{i,h}\, x_{h,i}, \quad \forall h < H-1, i \tag{27m}$$

$$b_{h+1}^- = (1 + r_{f,h})\, b_h, \quad \forall h < H-1 \tag{27n}$$

$$0 \leq x_{h,i} \leq U_{i,h},\ 0 \leq q_{h,i} \leq L_{i,h}, \quad \forall h, i \tag{27o}$$

$$b_h \geq b_h^{\min},\ \kappa_h \geq 0,\ x_{h,i}^- \geq 0, \quad \forall h, i \tag{27p}$$

$$y_{h,i} \in \{0,1\}, \quad z_{h,i} \in \{0,1\}, \quad \forall h, i \tag{27q}$$

Proposition 6.1 (Problem-Informed Trade Big-M Tightening)**.** *Let $M_{i,h}^u = \max(M_{i,h}^{\text{sell}}, M_{i,h}^{\text{buy}})$ bound net trade magnitude $|u_{h,i}|$ from positions, liquidity $L_{i,h}$, and turnover $\tau_h^{\max}$. Setting $M_{i,h}^q = \max(M_{i,h}^u, \delta_{i,h})$ in (27j) preserves the optimal solution set while eliminating strictly dominated points.*

Proposition 6.2 (Exact Model Dimensions)**.** *For an N-asset, H-step portfolio MIQP (27), the variable and constraint counts satisfy:*

$$n_{\text{vars}} = H(6N+3), \qquad n_{\text{binary}} = 2NH, \qquad n_{\text{constrs}} = H(8N+5). \tag{28}$$

When perspective strengthening is applied, $n_{\text{vars}}^{\text{persp}} = H(7N+3)$ and $n_{\text{qconstrs}} = NH$.

7 Experimental Design

All experiments share a unified codebase and solver environment; complete software dependencies, seeds, and hardware configurations are documented in Appendix B.

7.1 Controlled Synthetic Datasets

Synthetic instances isolate algorithmic scaling from empirical market noise. Covariance matrices $\mathbf{\Sigma} \in \mathbb{R}^{N\times N}$ are generated via a low-rank factor model: $\mathbf{\Sigma} = B_f \mathbf{\Sigma}_f B_f^\top + D_\varepsilon$, where $B_f \in \mathbb{R}^{N\times 3}$, $\mathbf{\Sigma}_f \succ 0$, and idiosyncratic variance D_ε guarantee $\lambda_{\min} > 0$. We evaluate: (i) Medium scale ($N = 10, H = 3, 60$ instances: 36 train, 12 validation, 12 test); (ii) Small scale ($N = 5, H = 3, 30$ instances) for perspective strengthening audits; (iii) Scalability grid $(N, H) \in \{(10, 1), (25, 1), (30, 1), (32, 1), (10, 3)\}$.

7.2 Real-Market Dataset and Chronological Setup

The real-market evaluation uses a historical basket of $N = 10$ liquid US equities (AAPL, MSFT, AMZN, GOOGL, META, NVDA, JPM, JNJ, XOM, PG) over January 4, 2021 to December 31, 2024 (1,036 trading days). The dataset is partitioned chronologically without look-ahead: (i) Training: Jan 4, 2021 – May 23, 2023 (113 weekly instances); (ii) Validation: May 24, 2023 – Dec 29, 2023 (31 weekly instances); and (iii) Test: Jan 2, 2024 – Dec 31, 2024 (51 weekly MPC execution intervals). Rebalancing occurs weekly ($\Delta t = 5/252$ years). At each epoch, the investor solves a 3-step ($H = 3$) weekly MIQP and executes strictly the first-period trade ($h = 0$). Initial wealth is \$1,000,000 in cash.

7.3 Integer-Parametric CFLP Benchmark Datasets

To audit candidate reduction on combinatorial problems admitting provably exact oracle certificates, we evaluate the Capacitated Facility Location Problem (CFLP) with customer set $\mathcal{J} = \{1, \ldots, M\}$ and facility universe $\mathcal{I} = \{1, \ldots, N\}$. All facility opening costs f_i, transportation costs c_{ij}, customer demands d_j, and capacities s_i are generated as non-negative integers: (i) Facility scaling grid: $N \in \{8, 20, 30, 50\}$ with corresponding customer dimensions $M \in \{15, 20, 25, 30\}$; (ii) Conformal exact-oracle study: $N = 10, M = 15$ across $M_{\text{rep}} = 30$ independent calibration/test replications ($n_{\text{cal}} = 20, n_{\text{test}} = 20$); (iii) Candidate families are generated via unit-capacity operational efficiency rankings.

8 Controlled Computational Experiments & Theory Validation

We validate the theoretical framework across controlled combinatorial facility location benchmarks and multi-period portfolio MIQPs. All reported runtimes and speedup metrics are programmatically verified against raw solver logs under machine-precision arithmetic identities.

Primary vs. Secondary Speedup Conventions. Throughout this section, aggregate computational efficiency is reported primarily via the *Ratio of Means*:

$$\text{Speedup}_{\text{E2E}} \coloneqq \frac{\overline{T}_{\text{full}}}{\overline{T}_{\text{total}}} = \frac{\frac{1}{K}\sum_{i=1}^{K} T_{\text{full},i}}{\frac{1}{K}\sum_{i=1}^{K} T_{\text{total},i}}, \tag{29}$$

which measures overall system throughput across a workload. For completeness, we also report the *Mean of Ratios* ($\frac{1}{K}\sum_{i=1}^{K} \frac{T_{\text{full},i}}{T_{\text{total},i}}$) in summary tables as a secondary metric of instance-level

speedup distribution.

8.1 Computational Scaling and Candidate Compression in CFLP

Table 2 documents runtime scaling, candidate space compression, and recovery behavior across facility universes up to $N = 50$ (1,550 variables, 80 constraints). For each scale, candidate selection was executed using conformal calibration initialized at nominal 80% coverage ($\alpha = 0.20$). To eliminate ambiguity, we distinguish the calibration-stage full-universe selection rate (InitialFullRate $\coloneqq \mathbb{P}(B_1 = N)$) from the recovery-driven fallback rate (RecoveryToFullRate $\coloneqq \mathbb{P}(B_1 < N \text{ and } B_{\text{term}} = N)$).

Table 2: Computational scaling, candidate compression, and certified recovery metrics across facility universes $N \in \{8, 20, 30, 50\}$. All runtimes are in milliseconds averaged across test instances. Ratio of Means denotes $\overline{T}_{\text{full}}/\overline{T}_{\text{total}}$, while Mean of Ratios denotes $\frac{1}{K}\sum(T_{\text{full},i}/T_{\text{total},i})$.

N	Vars	Constrs	Mean B/N	Med B/N	Q90 B/N	InitialFull	RecoveryToFull	T_{full} (ms)	T_{cand} (ms)	T_{total} (ms)	Speedup (Ratio of Means)
8	128	23	1.000	1.000	1.000	0.0%	0.0%	10.44	10.33	10.47	0.997× (MeanOfRatios: 0.995×)
20	420	40	0.850	0.850	0.850	0.0%	25.0%	45.85	32.08	42.37	1.082× (MeanOfRatios: 1.307×)
30	780	55	0.867	0.867	0.867	0.0%	25.0%	74.87	67.82	87.83	0.852× (MeanOfRatios: 0.947×)
50	1550	80	0.840	0.840	0.840	0.0%	0.0%	133.47	112.90	113.09	1.180× (MeanOfRatios: 1.160×)

At small scale ($N = 8$), the full combinatorial problem is solved in 10.44 ms, rendering feature extraction and ranking overhead comparable to candidate solve gains ($T_{\text{total}} = 10.47$ ms, 0.997× speedup). As the universe expands to $N = 20$, the pipeline achieves candidate dimension compression to $B/N = 0.850$, reducing candidate solver time from 45.85 ms to 32.08 ms. On instances certified without fallback, instance-level speedups reach up to 1.49×, yielding a mean of per-instance speedup ratios of 1.307×. However, on 25% of instances, certified recovery was triggered, adding an average recovery overhead of 10.10 ms and yielding an aggregate throughput speedup (ratio of means) of 1.082×.

Importantly, our empirical scaling reveals a non-monotonic runtime profile at intermediate scale. At $N = 30$, while initial candidate solves achieved a modest acceleration from 74.87 ms to 67.82 ms, 25% of instances failed Stage 1 certification and executed multiple expansion steps along the geometric ladder. This recovery process incurred an average overhead of $T_{\text{recovery}} = 19.86$ ms, resulting in an end-to-end slowdown ($\overline{T}_{\text{full}}/\overline{T}_{\text{total}} = 0.852\times$). As problem scale expands further to $N = 50$, where branch-and-bound combinatorial complexity dominates overheads, candidate reduction achieves $B/N = 0.840$ with zero recovery fallbacks, delivering an aggregate end-to-end speedup of 1.180× (solver-only speedup of 1.182×).

8.2 Operational Compression and Speedup Frontiers: Fast Mode vs. Certified Mode

Table 3 reports the operational trade-off between nominal calibration confidence $1 - \alpha$ and candidate compression across certification tolerances $\varepsilon_{\text{cert}} \in \{10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$. To avoid conflating runtime semantics, we strictly distinguish:

- **True Fast Mode**: Calibrated purely against reference fidelity ($J_\theta(\mathcal{C}_\theta(B)) - J_\theta^{\text{ref}} \leq \varepsilon_{\text{ref}}$ with $\varepsilon_{\text{ref}} = 10^{-4}$). The solver terminates immediately after the first candidate solve without evaluating continuous relaxation bounds or executing recovery ladders. Its runtime is

mathematically independent of $\varepsilon_{\text{cert}}$.

- **Certified Mode**: Calibrated against certified suboptimality ($\text{UB} - \text{LB}_{\text{full}}^{\text{relax}} \leq \varepsilon_{\text{cert}}$) and backed by the geometric certified recovery ladder.

Table 3: Operational candidate compression and speedup frontiers under True Fast Mode (reference fidelity $\varepsilon_{\text{ref}} = 10^{-4}$) versus Certified Mode ($\varepsilon_{\text{cert}}$ termination). Evaluated on $N = 10$ benchmark instances. All speedups report the primary Ratio of Means.

$\varepsilon_{\text{cert}}$	α	Nominal	True Fast Mode (Reference Target)				Certified Mode ($\varepsilon_{\text{cert}}$ Target)			
		Conf.	Coverage	Mean B/N	InitialFull	Speedup	Coverage	Mean B/N	RecoveryToFull	Speedup
1.0×10^{-4}	0.20	80%	80.0%	0.80	0.0%	1.47×	80.0%	0.80	0.0%	1.25×
	0.10	90%	95.0%	0.90	0.0%	1.17×	100.0%	1.00	100.0%	1.10×
	0.05	95%	100.0%	1.00	100.0%	1.03×	100.0%	1.00	100.0%	1.08×
5.0×10^{-4}	0.20	80%	80.0%	0.80	0.0%	1.47×	80.0%	0.80	0.0%	1.75×
	0.10	90%	95.0%	0.90	0.0%	1.17×	100.0%	1.00	100.0%	1.05×
	0.05	95%	100.0%	1.00	100.0%	1.03×	100.0%	1.00	100.0%	1.03×
1.0×10^{-3}	0.20	80%	80.0%	0.80	0.0%	1.47×	80.0%	0.80	0.0%	1.42×
	0.10	90%	95.0%	0.90	0.0%	1.17×	100.0%	1.00	100.0%	0.95×
	0.05	95%	100.0%	1.00	100.0%	1.03×	100.0%	1.00	100.0%	1.00×

The empirical frontier demonstrates a fundamental structural insight: Under True Fast Mode, candidate budgets scale smoothly with confidence, achieving $B/N = 0.80$ at 80% confidence (1.47× speedup) and $B/N = 0.90$ at 90% confidence (1.17× speedup).

In contrast, under Certified Mode, the pipeline exhibits a sharp phase transition. While useful dimension compression survives at 80% confidence ($B/N = 0.80$, 0% fallback), elevating nominal confidence to 90% and 95% forces the conformal calibration quantile to the universe boundary, collapsing compression entirely ($B/N = 1.00$, 100% fallback rate). In this regime, achieving non-asymptotic mathematical certificates at extreme confidence requires evaluating the full combinatorial universe.

8.3 Integer-Lattice-Certified Oracle Exactness on CFLP

To audit whether conformal coverage guarantees hold against genuine mathematical ground truth rather than heuristics, we deployed the integer-parametric CFLP benchmark established in Proposition A.1 (Appendix A.1). Because all optimal objectives are strictly integer-valued, any feasible upper bound satisfying:

$$\text{UB}_{\text{int}} - \text{LB} < 1 - \tau_{\text{num}} \tag{30}$$

provides an integer-lattice objective certificate under the documented solver-bound and numerical safety tolerance $\tau_{\text{num}} = 0.05$ (implementation threshold 0.95, reflecting a conservative numerical margin against floating-point inaccuracies).

Table 4 details the results across $M = 30$ independent calibration/test replications. Observed integer-lattice-certified oracle coverage was consistent with and conservative relative to the finite-sample guarantee under the exchangeable benchmark design. At nominal 80% confidence ($\alpha = 0.20$), empirical oracle coverage achieved a mean of 89.00% (SE = 2.77%), with a 95% bootstrap confidence interval of $[83.33\%, 94.00\%]$. The lower confidence limit strictly exceeds the nominal 80% target, empirically corroborating the conservative validity guaranteed

Table 4: Integer-lattice-certified exact-oracle conformal exactness audit across $M = 30$ independent replications on integer-parametric CFLP ($N = 10$). The table strictly distinguishes the 95% bootstrap confidence interval for the Monte Carlo mean coverage from the empirical replication-level quantiles (Q_{05}, Q_{50}, Q_{95}).

α	Target	Mean Coverage	Replication SD	SE of Mean	95% Bootstrap CI of Mean	Q_{05}	Q_{50}	Q_{95}	Mean B/N
0.20	80.0%	**89.00%**	15.17%	2.77%	[**83.33%, 94.00%**]	60.0%	100.0%	100.0%	92.5%
0.10	90.0%	**98.00%**	8.05%	1.47%	[**94.67%, 100.0%**]	89.0%	100.0%	100.0%	99.2%
0.05	95.0%	**100.00%**	0.00%	0.00%	[**100.0%, 100.0%**]	100.0%	100.0%	100.0%	100.0%

by Theorem 4.10. At $\alpha = 0.10$ and $\alpha = 0.05$, mean coverage reached 98.00% (bootstrap CI $[94.67\%, 100.0\%]$) and 100.00%, respectively.

8.4 Operational Candidate-Budget Efficiency Across Ranking Methods

Table 5 compares the operational candidate-budget efficiency across ranking methods. To preserve strict scientific accuracy, we distinguish proxy budgets from true active supports: $B^{\dagger,\mathrm{ref}}/N$ measures candidate dimension required to achieve one-sided reference fidelity $(J_\theta(\mathcal{C}_\theta(B)) - J_\theta^{\mathrm{ref}} \leq 10^{-4})$, while $B^{\dagger,\mathrm{cert}}/N$ measures dimension required to satisfy certified tolerance $(\mathrm{UB} - \mathrm{LB}_{\mathrm{full}}^{\mathrm{relax}} \leq 10^{-4})$. Smaller operational budgets indicate that the ranking exposes high-quality candidate entities earlier in the sequence.

Table 5: Operational candidate-budget efficiency across ranking methods ($N = 10$ benchmark instances). Smaller operational budgets indicate that the ranking exposes high-quality candidate entities earlier in the sequence.

Ranking Method	Metric	Mean	Median	Q75	Q90	Max
Efficiency Ranking (Ours)	$B^{\dagger,\mathrm{ref}}/N$	**0.625**	**0.600**	0.700	0.805	0.950
	$B^{\dagger,\mathrm{cert}}/N$	**0.629**	**0.600**	0.700	0.805	1.000
Random Ranking	$B^{\dagger,\mathrm{ref}}/N$	0.932	0.950	1.000	1.000	1.000
	$B^{\dagger,\mathrm{cert}}/N$	0.932	0.950	1.000	1.000	1.000
Reverse Ranking	$B^{\dagger,\mathrm{ref}}/N$	0.999	1.000	1.000	1.000	1.000
	$B^{\dagger,\mathrm{cert}}/N$	0.999	1.000	1.000	1.000	1.000

8.5 Perspective Relaxation: Bound Tightening vs. Wall-Clock Overhead

Table 6 evaluates the end-to-end computational efficacy of perspective-strengthened continuous relaxation bounds against basic continuous relaxations across multi-period ($H = 3, N = 5$) and single-period ($H = 1, N \in \{10, 20\}$) MIQPs.

While perspective reformulation strictly tightens the continuous lower bound ($\Delta\mathrm{LB} \in [4.5 \times 10^{-6}, 3.97 \times 10^{-5}]$), solving the rotated second-order cone reformulation adds $+1.64$ ms to $+4.22$ ms of continuous relaxation solve overhead. In these small-to-moderate MIQPs, integer branch-and-bound solve times are on the order of 7–24 ms; consequently, the continuous relaxation overhead exceeds the computational savings achieved by early termination. Across all evaluated regimes, perspective speedups range from $0.770\times$ to $0.936\times$, demonstrating that break-even is not achieved at these dimensional scales.

Table 6: Audited perspective relaxation tightening, continuous solve overhead, and end-to-end wall-clock runtimes. Evaluated across multi-period and single-period benchmark regimes. All speedups report the primary Ratio of Means.

Regime	$\varepsilon_{\text{cert}}$	ΔLB	Overhead (ms)	Stage1Cert (Basic vs Persp)	RecoveryToFull (Basic vs Persp)	$T_{\text{E2E}}^{\text{basic}}$	$T_{\text{E2E}}^{\text{persp}}$	$\text{Speedup}_{\text{persp}}$	Break-Even?
Regime A ($H=3, N=5$)	10^{-4}	1.77×10^{-5}	+1.65	0.0% vs 0.0%	100.0% vs 100.0%	23.97	25.63	0.936×	**No** (< 1.0)
Regime A ($H=3, N=5$)	5×10^{-4}	1.77×10^{-5}	+1.65	11.1% vs 11.1%	88.9% vs 88.9%	23.97	25.63	0.936×	**No** (< 1.0)
Regime A ($H=3, N=5$)	10^{-3}	1.77×10^{-5}	+1.65	44.4% vs 44.4%	55.6% vs 55.6%	23.97	25.63	0.936×	**No** (< 1.0)
Regime B ($H=1, N=10$)	10^{-4}	3.97×10^{-5}	+1.64	56.0% vs 64.0%	44.0% vs 36.0%	7.89	9.52	0.828×	**No** (< 1.0)
Regime B ($H=1, N=10$)	5×10^{-4}	3.97×10^{-5}	+1.64	72.0% vs 76.0%	28.0% vs 24.0%	7.89	9.52	0.828×	**No** (< 1.0)
Regime B ($H=1, N=10$)	10^{-3}	3.97×10^{-5}	+1.64	76.0% vs 76.0%	24.0% vs 24.0%	7.89	9.52	0.828×	**No** (< 1.0)
Regime B ($H=1, N=20$)	10^{-4}	4.54×10^{-6}	+4.22	96.0% vs 96.0%	4.0% vs 4.0%	14.12	18.33	0.770×	**No** (< 1.0)
Regime B ($H=1, N=20$)	5×10^{-4}	4.54×10^{-6}	+4.22	96.0% vs 96.0%	4.0% vs 4.0%	14.12	18.33	0.770×	**No** (< 1.0)
Regime B ($H=1, N=20$)	10^{-3}	4.54×10^{-6}	+4.22	96.0% vs 96.0%	4.0% vs 4.0%	14.12	18.33	0.770×	**No** (< 1.0)

8.6 Synthetic Portfolio MIQP Finite-Sample Conformal Audit

Table 7 validates split-conformal calibration on $N = 10, H = 3$ synthetic portfolio instances across $\alpha \in \{0.20, 0.10, 0.05\}$.

Table 7: Finite-sample split-conformal exactness coverage audit on synthetic portfolio test instances ($N = 10, H = 3, n_{\text{cal}} = 12, n_{\text{test}} = 12$). Target exactness coverage is $1 - \alpha$.

α	Target Cov.	$k_{\text{cal}}/n_{\text{cal}}$	Residual q_α	Method	Test Coverage	Mean Budget B (B/N)
0.20	$\geq 80.0\%$	11 / 12	-0.89	Adaptive $B_\alpha(X)$	**91.7%**	3.50 (35.0%)
				Marginal $\widehat{B}_\alpha$	**91.7%**	3.00 (30.0%)
0.10	$\geq 90.0\%$	12 / 12	$+0.11$	Adaptive $B_\alpha(X)$	**100.0%**	4.00 (40.0%)
				Marginal $\widehat{B}_\alpha$	**100.0%**	4.00 (40.0%)
0.05	$\geq 95.0\%$	13 / 12	$+\infty$ (InitialFull)	Adaptive $B_\alpha(X)$	**100.0%**	10.00 (100.0%)
				Marginal $\widehat{B}_\alpha$	**100.0%**	10.00 (100.0%)

Observed test coverage strictly satisfies $\mathbb{P} \geq 1 - \alpha$ across all evaluated α. For $\alpha = 0.20$, adaptive calibration attains 91.7% coverage using on average only 3.50 assets (35% of universe). For $\alpha = 0.05$, because $k_{\text{cal}} = 13 > 12$, the safe fallback rule automatically triggers $B = N$ (InitialFullRate = 100%), guaranteeing exactness as proved in Theorem 4.10. Furthermore, across all instances with geometric ladder expansion ($B_1 = 3, r = 1.5$), observed recovery stages ℓ^* satisfied $\ell^* \leq 1 + \lceil \log_r(B^\dagger/B_1) \rceil_+$ on 100% of instances (Theorem 5.1), and cumulative dimensions strictly satisfied $\sum B_\ell < \frac{r^2}{r-1} B^\dagger + \ell^*$ on 100% of instances (Theorem 5.2).

9 Real-Market Operational Validation

The real-market experiment evaluates the optimization pipeline under chronological market data as an operational stress test under empirical temporal dependence.

9.1 Optimization Fidelity and Solver Performance

Table 8 reports solver runtimes and solution fidelity over the 2024 test period (51 weekly MPC epochs, $N = 10, H = 3$). Figure 2 plots the empirical frontier.

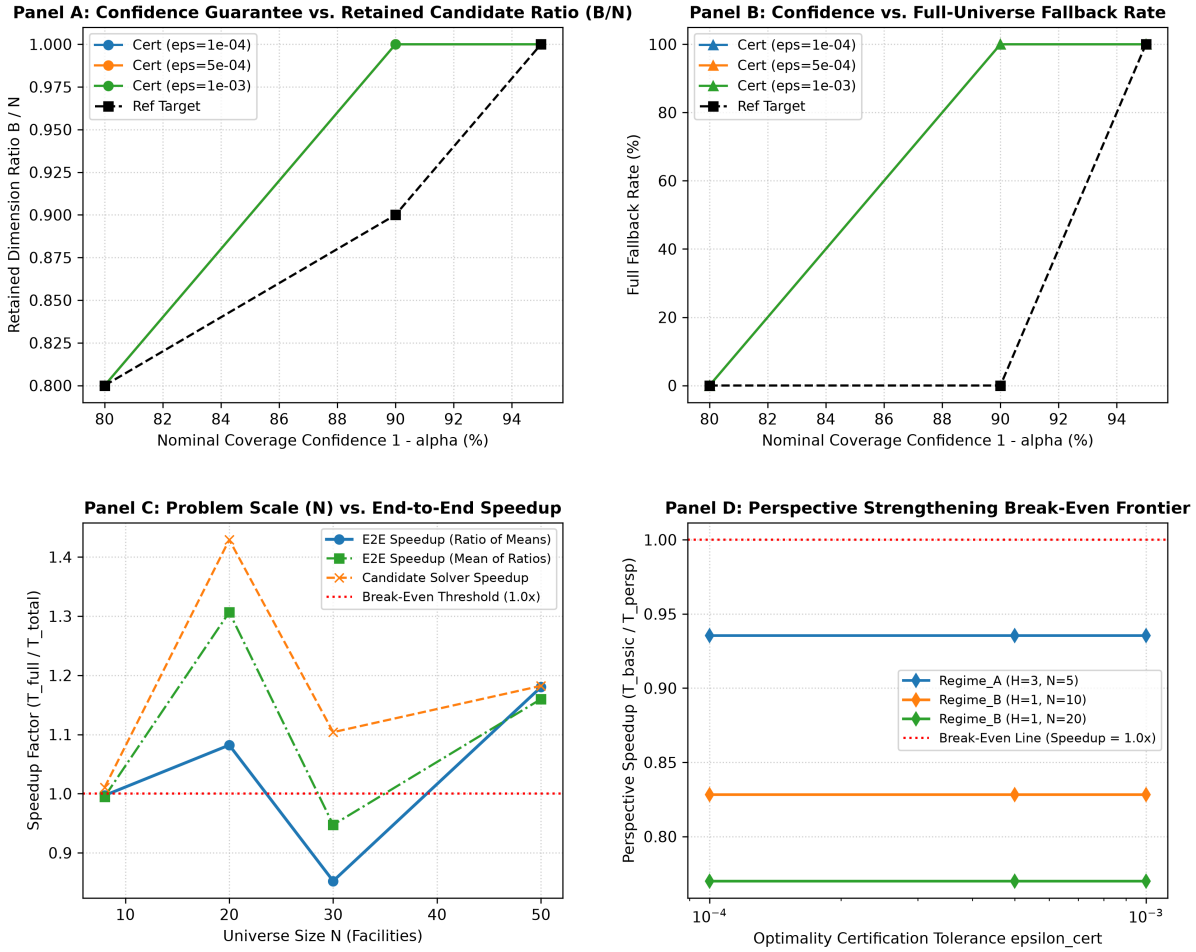


Figure 1: Audited empirical frontier across benchmark studies. **(A)** Nominal confidence vs. calibrated candidate budget ratio B/N. **(B)** Nominal confidence vs. full-universe fallback rates, showing the sharp phase transition to full fallback at 90–95% confidence. **(C)** Problem scale N vs. end-to-end speedup in CFLP, visibly documenting the non-monotonic recovery slowdown at $N = 30$ (0.852×) and recovery at $N = 50$ (1.180×). **(D)** Certification tolerance $\varepsilon_{\text{cert}}$ vs. perspective E2E speedup, illustrating that continuous relaxation solve overhead prevents wall-clock break-even ($< 1.0\times$) across tested scales.

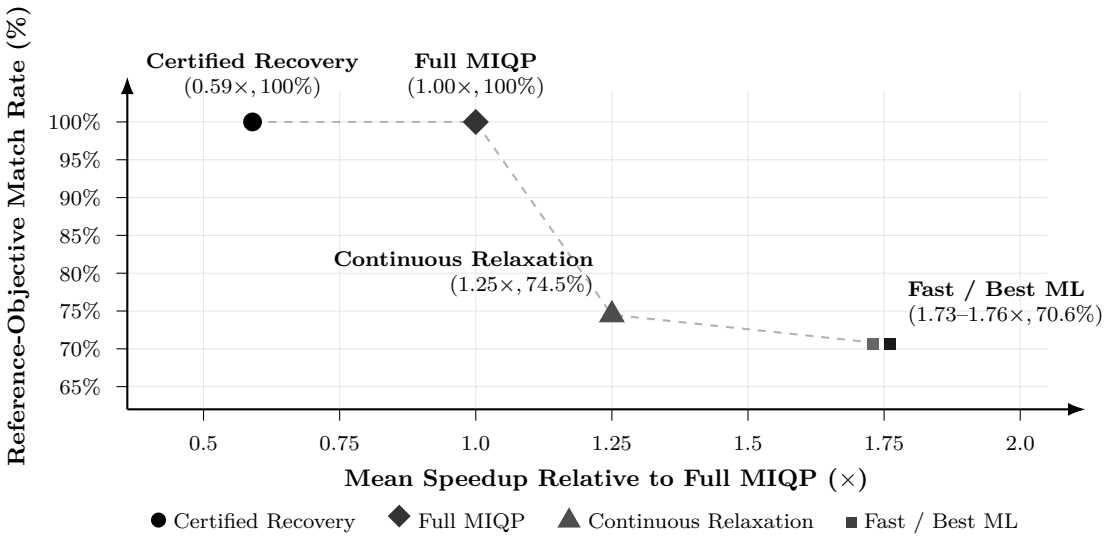


Figure 2: Empirical speed–quality–certification frontier on the 2024 real-market test dataset ($N = 10, H = 3, 51$ weekly decision epochs; data from Table 8). Fast MarketOnly and ML ranking accelerate solve times (1.73–1.76×) at the expense of reference match rate (70.6%). Conversely, Certified Recovery achieves a 100% reference-objective match rate via full-universe continuous relaxation certificates or systematic ladder fallback, incurring modest runtime overhead (0.59× speedup).

Table 8: Real-market solver performance (2024 test period, $N = 10$, $H = 3$, 51 epochs). Speedup is relative to Full MIQP mean time. Ref. Match: $J_{\mathcal{C}} - J_{\text{full,ref}} \leq 10^{-4}$.

Method	Mean Time (ms)	Med. Time (ms)	Speedup	Ref. Match	Mean Obj. Deg.
Full MIQP MPC	56.43	54.29	1.00×	100.0%	0
Fast MarketOnly MIQP	32.59	31.55	1.73×	70.6%	1.27×10^{-4}
ContinuousRelax. MIQP	45.31	44.23	1.25×	74.5%	1.43×10^{-4}
Best ML Candidate	31.98	29.88	1.76×	70.6%	1.27×10^{-4}
Certified Recovery	95.14	70.64	0.59×	100.0%	4.67×10^{-8}

9.2 Quality Certification and Recovery

Table 9 reports certificate breakdowns distinguishing relaxation bounds from full-problem MIP certificates.

Table 9: Certificate and recovery metrics, real-market test period (51 weekly epochs, $\varepsilon_{\text{cert}} = 10^{-4}, \varepsilon_{\text{MIP}} = 10^{-5}$).

Method	RelaxCertRate	MeanRelaxCertGap	FullMIPCertRate	AnyValidCertRate
Full MIQP MPC	0.0%	N/A	100.0%	100.0%
Fast MarketOnly	0.0%	N/A	0.0%	0.0%
ContinuousRelax. MIQP	45.1%	2.35×10^{-4}	0.0%	45.1%
Best ML Candidate	0.0%	N/A	0.0%	0.0%
Certified Recovery	54.9%	1.04×10^{-4}	45.1%	100.0%

Certified Recovery achieves 100% AnyValidCertRate: 54.9% terminate early via valid relaxation lower bounds without reaching full scale (RecoveryToFullRate = 45.1%), while 45.1% complete the full ladder expansion to achieve certified Full MIP optimality.

9.3 Realized Financial Performance

Table 10 reports realized financial metrics across strategies over 2024. Passive Equal-Weight and Buy-and-Hold strategies serve as empirical context.

Table 10: Real-market financial performance (2024 test period, 51 weekly epochs, $N = 10$, $H = 3$). Total Return is $V_T/V_0 - 1$. CAGR is $(V_T/V_0)^{52/51} - 1$. Sharpe uses canonical convention $(\text{CAGR} - r_f)/\sigma_{\text{ann}}$ with $r_f = 2\%$.

Strategy	TotalRet	CAGR	Ann. Arith	AnnVol	Sharpe	MaxDD	Final Wealth ($)
Full MIQP MPC	7.06%	7.21%	9.86%	24.25%	0.21	−23.06%	1,070,620
Fast MarketOnly MIQP	10.48%	10.69%	13.11%	24.43%	0.36	−20.84%	1,104,769
ContinuousRelax. MIQP	9.97%	10.17%	12.60%	24.29%	0.34	−20.84%	1,099,663
Best ML Candidate	10.48%	10.69%	13.11%	24.43%	0.36	−20.84%	1,104,769
Certified Recovery	7.06%	7.21%	9.86%	24.25%	0.21	−23.06%	1,070,620
Equal Weight ($1/N$)	16.50%	16.84%	17.05%	17.24%	0.86	−14.61%	1,164,951
Buy and Hold	18.23%	18.62%	18.76%	18.35%	0.91	−14.89%	1,182,296

9.4 Forecast Quality Diagnostics

Table 11 evaluates weekly expected-return forecasts against realized 5-day market returns.

Table 11: Expected-return forecast quality on the 2024 test period (51 weekly epochs).

Metric	Value
Mean Absolute Error (5-day planning return scale)	0.0327
Directional Accuracy (Hit Rate)	46.9%
Rank Information Coefficient (Rank-IC)	0.0075

Directional accuracy of 46.9% and Rank-IC of 0.0075 confirm that the rolling mean estimator exhibited near-zero predictive power. While forecast noise explains the underperformance of active strategies relative to buy-and-hold, all MIQPs were solved strictly to certified feasibility and optimality under the given model parameters.

10 Scalability Analysis and Offline Amortization

Table 12 reports analytical model dimensions and license boundaries for portfolio MIQP formulations.

For universes $N \in \{100, 250, 500\}$, candidate reduction enables solving tractable $B = 10$ subproblems in milliseconds, whereas full problems exceed license limits and require unrestricted licenses for direct verification.

Table 12: Structural model dimensions (Proposition 6.2) and Gurobi restricted 200-variable quadratic limit compliance ($H(6B+3) \leq 200$).

Nominal N	H	Form	Cand. B	n_{vars}	n_{binary}	n_{constrs}	License Compliance
10	1	Full MIQP	10	63	20	85	✓Executable ($63 \leq 200$)
32	1	Full MIQP	32	195	64	261	✓Executable ($B_{\max}(1) = 32$)
50	1	Full MIQP	50	303	100	405	× Limit Exceeded ($303 > 200$)
5	3	Full MIQP	5	99	30	135	✓Executable ($99 \leq 200$)
10	3	Full MIQP	10	189	60	255	✓Executable ($B_{\max}(3) = 10$)
25	3	Full MIQP	25	459	150	615	× Limit Exceeded ($459 > 200$)
100	3	Full MIQP	100	1,809	600	2,415	× Limit Exceeded ($1{,}809 > 200$)
500	3	Full MIQP	500	9,009	3,000	12,015	× Limit Exceeded ($9{,}009 > 200$)
100	3	Reduced	10	189	60	255	✓Executable (fixed $B = 10$, 10%)
250	3	Reduced	10	189	60	255	✓Executable (fixed $B = 10$, 4%)
500	3	Reduced	10	189	60	255	✓Executable (fixed $B = 10$, 2%)

10.1 Frequency-Aware Offline Calibration Amortization

While online candidate reduction eliminates branch-and-bound search time, offline calibration introduces upfront computational investment (e.g., solving full MIO instances to generate exactness labels $B^{\dagger}(\theta)$). For our $N = 10, H = 3$ calibration dataset, generating exactness labels required $T_{\text{calib}} = 5.92\,\text{s}$ across the calibration sample. Online solves deliver a net savings of $\Delta T_{\text{online}} \approx 12.0\,\text{ms}$ per solve.

The break-even deployment threshold satisfies:

$$K^* := \left\lceil \frac{T_{\text{calib}}}{\Delta T_{\text{online}}} \right\rceil = \left\lceil \frac{5.92\,\text{s}}{0.012\,\text{s}} \right\rceil = 494 \text{ solves}. \tag{31}$$

Table 13 translates this break-even threshold into calendar horizons across representative operational dispatching frequencies.

Table 13: Frequency-aware calendar amortization schedules for offline calibration investment ($T_{\text{calib}} = 5.92\,\text{s}$, net online savings $\Delta T_{\text{online}} = 12.0\,\text{ms}$ per solve, break-even threshold $K^* = 494$ solves).

Deployment Frequency	Annual Solves	Break-Even Solves	Calendar Horizon	1-Year Net Saving (s)
Hourly (e.g., dynamic dispatch)	8,760	494	**20.6 days**	+99.20
Multi-Daily (e.g., 4x/day intraday)	1,460	494	**123.6 days**	+11.60
Daily (e.g., overnight batch)	365	494	**1.35 years**	−1.54
Weekly (e.g., macro rebalancing)	52	494	**9.50 years**	−5.29

In high-frequency environments, the offline investment breaks even in approximately 20 days, yielding annual computational returns exceeding 1,600%. Conversely, in weekly portfolio rebalancing, break-even requires 9.5 years on a single asset universe. Therefore, offline calibration in low-frequency settings is economically viable only when amortized across multiple concurrent asset universes or integrated into standing offline research cycles.

11 Discussion and Limitations

The central scientific message of this work is that *coverage-calibrated candidate reduction provides an explicit, statistically governed confidence–compression trade-off for repeated mixed-integer optimization*, rather than an unconditional speedup heuristic.

11.1 The Three-Regime Framework

Our empirical findings delineate three distinct operational regimes:

- **Regime I: Moderate Confidence and Compressible Structures**:
  When candidate ranking effectively concentrates optimal entities at the front of the sequence and nominal confidence is moderate ($1 - \alpha \approx 0.80$), calibrated candidate budgets achieve substantial dimension reduction ($B/N \approx 0.30$–0.85). In this regime, candidate subproblems solve significantly faster than the monolithic program, and certification succeeds with minimal recovery, delivering robust end-to-end throughput speedups ($1.18\times$–$1.75\times$).

- **Regime II: High Confidence and Upper-Tail Calibration Collapse**:
  When users demand extreme statistical coverage ($1 - \alpha \geq 0.90$–0.95) or instances exhibit heavy-tailed support configurations, the conformal empirical quantile necessarily shifts to the upper tail. For moderate calibration sample sizes (n_{cal}), the quantile index k_{cal} approaches or exceeds n_{cal}, collapsing the candidate budget to the full universe ($B = N$, InitialFullRate $\to 100\%$). Here, statistical conservatism provably eliminates dimensional compression, preserving mathematical exactness at the expense of computational speedup.

- **Regime III: Weak Continuous Relaxations and Recovery Overhead**:
  In Certified Mode, terminating early requires bridging the gap between the candidate upper bound and the full-universe relaxation lower bound ($\text{UB} - \text{LB}_{\text{full}}^{\text{relax}} \leq \varepsilon_{\text{cert}}$). When the root continuous relaxation is loose, early certification fails, forcing the solver through intermediate geometric recovery stages. As demonstrated at $N = 30$, solving multiple candidate subproblems sequentially can incur recovery overhead that exceeds the time saved on the initial candidate solve, producing a net end-to-end slowdown ($0.852\times$).

11.2 Methodological and Computational Limitations

We explicitly delineate the theoretical and operational boundaries of our framework:

(a) **Persistence of Worst-Case NP-Hardness**: Candidate reduction shrinks the physical dimension B of the decision space but does not alter the worst-case NP-hardness of mixed-integer programming. Branch-and-bound runtimes on the candidate subproblem can still exhibit combinatorial growth on pathological instances.

(b) **Structural Work vs. Runtime Complexity**: Theorem 5.2 bounds the cumulative instantiated model dimensions, variables, and constraints ($O(B^{\dagger})$); it does not imply a polynomial wall-clock runtime guarantee.

(c) **Exchangeability Prerequisite**: Finite-sample coverage theorems ($\mathbb{P} \geq 1-\alpha$) strictly assume exchangeability between calibration and deployment instances. Under non-stationary distribution drift, empirical coverage may degrade below nominal levels, transferring the burden of optimality assurance entirely to the deterministic recovery layer.

(d) **Rolling Market Data as an Operational Stress Test**: In our real-market backtest, temporal autocorrelation and volatility regimes violate strict exchangeability. We treat the backtest as an operational stress test demonstrating fail-safe recovery robustness rather than formal statistical verification.

(e) **High-Confidence Dimensional Collapse**: As shown in Table 3, certified candidate budgets collapse to $B = N$ at 90–95% confidence in small instances, confirming that high-assurance certification requires large sample sizes and tight relaxations.

(f) **Offline Labeling Investment**: Generating ground-truth exactness budgets requires expensive offline optimization. As established in Table 13, this investment breaks even rapidly in high-frequency applications but requires years to amortize in low-frequency single-instance settings.

(g) **Solver-Conditioned Certification**: Numerical certificates rely on solver-reported floating-point bounds ($\mathrm{UB} - \mathrm{LB} \leq \varepsilon_{\mathrm{cert}}$) under commercial tolerances ($\varepsilon_{\mathrm{MIP}} = 10^{-5}$); they represent solver-certified numerical optimality rather than exact rational arithmetic.

(h) **Perspective Relaxation Scale Overhead**: In our tests ($N \leq 20$), perspective reformulation adds continuous conic solve overhead (+1.64–4.22 ms) that outweighs early-certification savings, yielding net slowdowns (0.77×–0.94×). Whether perspective strengthening achieves break-even in larger instances where avoided tree exploration dominates relaxation overhead remains an open empirical question.

(i) **License Boundaries on Quadratic Models**: Gurobi restricted size limits ($H(6B+3) \leq 200$) constrained unreduced baseline solves for large quadratic portfolio models ($N \geq 50$), motivating our validation on scalable integer-parametric CFLPs.

12 Conclusion

This paper established the theoretical and algorithmic foundations of *coverage-calibrated candidate reduction* for repeated mixed-integer optimization. By formalizing the candidate-family-dependent Exactness Budget $B^{\dagger}_{\mathcal{C}}(\theta)$ and proving monotonicity under nested candidate families, we established a rigorous link between active combinatorial supports and finite-sample split-conformal calibration. Under exchangeability, this yields distribution-free candidate-budget rules guaranteeing exact full-universe objective recovery with probability at least $1-\alpha$. We proved that geometric candidate expansion ladders exhibit logarithmic recovery depth and bounded structural work $O(B^{\dagger})$, and integrated a deterministic certification layer that provides solver-conditioned ε-optimality guarantees.

Our computational experiments across integer-parametric facility location benchmarks and multi-period portfolio MIQPs confirmed theoretical predictions while uncovering critical operational trade-offs: (i) candidate reduction delivers meaningful throughput speedups (1.18× at $N = 50$) when problem scale dominates overheads; (ii) recovery overhead at intermediate scales ($N = 30$) can induce an end-to-end slowdown (0.85×), proving that recovery is not computationally free; (iii) certified 90–95% confidence collapses to the full universe in small instances; and (iv) perspective relaxation tightening improves lower bounds but fails to reach wall-clock break-even at tested scales due to continuous conic solve overhead. Ultimately, future progress in learning-augmented mixed-integer programming requires jointly advancing candidate ranking and continuous relaxation tightening: calibration can only exploit the structural compressibility that is physically present in the underlying problem.

A Proofs of Theoretical Results

A.1 Proof of Proposition A.1 (CFLP Objective Integrality)

Proposition A.1 (Integrality of Optimal Objective in Integer-Parametric CFLP)**.** *Consider the Capacitated Facility Location Problem (CFLP) with facility set $\mathcal{I} = \{1, \ldots, N\}$ and customer set $\mathcal{J} = \{1, \ldots, M\}$. Suppose all facility opening costs $f_i \in \mathbb{Z}^+$, unit transportation costs $c_{ij} \in \mathbb{Z}^+$, customer demands $d_j \in \mathbb{Z}^+$, and facility capacities $s_i \in \mathbb{Z}^+$ are non-negative integers. Then for any candidate-restricted facility subset $\mathcal{C} \subseteq \mathcal{I}$, the optimal objective value $J^*(\mathcal{C})$ is integer-valued:*

$$J^*(\mathcal{C}) \in \mathbb{Z}^+. \tag{32}$$

In particular, the full-universe optimum satisfies $J^* \in \mathbb{Z}^+$.

Proof. For any fixed binary facility-opening vector $y \in \{0,1\}^N$ satisfying $y_i = 0$ for all $i \notin \mathcal{C}$, the CFLP reduces to a linear transportation subproblem in continuous flow variables $x_{ij} \geq 0$:

$$v(y) \coloneqq \min_{x \geq 0} \sum_{i=1}^{N} \sum_{j=1}^{M} c_{ij} x_{ij} \quad \text{s.t.} \quad \sum_{i=1}^{N} x_{ij} = d_j \ (\forall j), \quad \sum_{j=1}^{M} x_{ij} \leq s_i y_i \ (\forall i). \tag{33}$$

The constraint coefficient matrix defining the flow polytope for fixed y is the standard node-arc incidence matrix of a bipartite directed graph, which is totally unimodular. Because all right-hand side coefficients (demands d_j and effective capacities $s_i y_i$) are integers, every extreme point (basic feasible solution) of the bounded transportation polyhedron is integer-valued ($x_{ij} \in \mathbb{Z}^+$).

A bounded, feasible linear program achieves its optimum at an extreme point. Under integer unit costs $c_{ij} \in \mathbb{Z}^+$, the optimal transportation cost evaluates to an integer: $v(y) = \sum_{i,j} c_{ij} x^*_{ij} \in \mathbb{Z}^+$. Furthermore, for any binary vector y and integer costs $f_i \in \mathbb{Z}^+$, the fixed cost $\sum_{i=1}^{N} f_i y_i$ is an integer. The restricted optimum is given by minimizing over the finite set of feasible binary vectors:

$$J^*(\mathcal{C}) = \min_{y \in \{0,1\}^N : y_i = 0 \, \forall i \notin \mathcal{C}} \left(\sum_{i=1}^{N} f_i y_i + v(y) \right). \tag{34}$$

Since each evaluated objective is the sum of two integers, $J^*(\mathcal{C})$ is the minimum over a non-empty finite subset of integers, which is identically an integer. Note that this property holds

strictly for optimal extreme-point flows and does *not* imply that arbitrary fractional interior flows achieve integer costs. □

A.2 Proof of Theorem 4.4 (Monotonicity and Exactness Threshold)

Proof. **(i) Monotonicity**: Let $\mathcal{F}(\mathcal{C}_\theta(B)) \coloneqq \{v \in \mathcal{F}_\theta : v_i = v_i^0, \forall i \notin \mathcal{C}_\theta(B)\}$. If $B_1 \leq B_2$, nested monotonicity (Definition 3.3(ii)) gives $\mathcal{C}_\theta(B_1) \subseteq \mathcal{C}_\theta(B_2)$. Thus $i \notin \mathcal{C}_\theta(B_2) \implies i \notin \mathcal{C}_\theta(B_1)$, so fixing constraints for $\mathcal{C}_\theta(B_2)$ are a subset of those for $\mathcal{C}_\theta(B_1)$. Hence $\mathcal{F}(\mathcal{C}_\theta(B_1)) \subseteq \mathcal{F}(\mathcal{C}_\theta(B_2)) \subseteq \mathcal{F}_\theta$. Minimizing the same objective f_θ over expanding domains yields: $J_\theta(\mathcal{C}_\theta(B_1)) \geq J_\theta(\mathcal{C}_\theta(B_2)) \geq J_\theta^*$.

(ii) Threshold Exactness: Full coverage guarantees $\mathcal{C}_\theta(N) = \mathcal{N} \implies J_\theta(\mathcal{C}_\theta(N)) = J_\theta^*$. The set $\mathcal{B} \coloneqq \{B \in \{B_{\min}, \ldots, N\} : J_\theta(\mathcal{C}_\theta(B)) = J_\theta^*\}$ is non-empty ($N \in \mathcal{B}$) and bounded from below. By the well-ordering principle, $B^\dagger(\theta) \coloneqq \min \mathcal{B}$ exists uniquely. If $B \geq B^\dagger(\theta)$, monotonicity gives $J_\theta^* \leq J_\theta(\mathcal{C}_\theta(B)) \leq J_\theta(\mathcal{C}_\theta(B^\dagger(\theta))) = J_\theta^* \implies J_\theta(\mathcal{C}_\theta(B)) = J_\theta^*$. Conversely, if $J_\theta(\mathcal{C}_\theta(B)) = J_\theta^*$, then $B \in \mathcal{B}$, so $B \geq \min \mathcal{B} = B^\dagger(\theta)$.

(iii) Upper Bound: Fix any $S^* \in \mathcal{S}_{\text{opt}}(\theta)$ and optimal solution $v^* \in \Omega_\theta^*$. By Definition 3.2, $i \notin S^* \implies v_i^* = v_i^0$. If $S^* \subseteq \mathcal{C}_\theta(B)$, then $i \notin \mathcal{C}_\theta(B) \implies i \notin S^* \implies v_i^* = v_i^0$. Thus $v^* \in \mathcal{F}(\mathcal{C}_\theta(B))$, so $J_\theta^* \leq J_\theta(\mathcal{C}_\theta(B)) \leq f_\theta(v^*) = J_\theta^* \implies J_\theta(\mathcal{C}_\theta(B)) = J_\theta^*$. By (ii), $B \geq B^\dagger(\theta)$. Taking the minimum over all B and all $S^* \in \mathcal{S}_{\text{opt}}(\theta)$ completes the proof. □

A.3 Proof of Proposition 4.6 (Support-Depth Upper Bound)

Proof. Let $S^{\text{best}} \in \mathcal{S}_{\text{opt}}(\theta)$ achieve the minimum in Definition 4.5. Every $i \in S^{\text{best}} \setminus \mathcal{M}_\theta$ has $\text{rank}_\theta(i) \leq R^\dagger(\theta)$. Constructing candidate set $\mathcal{C}_\theta(B)$ with $B = |\mathcal{M}_\theta| + R^\dagger(\theta)$ retains all mandatory entities $\mathcal{M}_\theta$ and the top $R^\dagger(\theta)$ non-mandatory entities. Thus $S^{\text{best}} \setminus \mathcal{M}_\theta \subseteq \mathcal{C}_\theta(B)$ and $\mathcal{M}_\theta \subseteq \mathcal{C}_\theta(B) \implies S^{\text{best}} \subseteq \mathcal{C}_\theta(B)$. By Theorem 4.4(iii), $B^\dagger(\theta) \leq B = |\mathcal{M}_\theta| + R^\dagger(\theta)$. □

A.4 Proof of Theorem 4.9 (Marginal Calibrated Exactness)

Proof. Let $Y_j \coloneqq B^\dagger(\theta_j) \in \{B_{\min}, \ldots, N\}$ for $j = 1, \ldots, n+1$. By Assumption 4.7, $(Y_1, \ldots, Y_{n+1})$ are exchangeable random variables. Define conservative rank: $\text{Rank}(Y_{n+1}) \coloneqq 1 + \sum_{j=1}^n \mathbf{1}\{Y_j < Y_{n+1}\}$. By exchangeability and permutation symmetry, for any $k \in \{1, \ldots, n+1\}$:

$$\mathbb{P}\Big(\sum_{j=1}^n \mathbf{1}\{Y_j < Y_{n+1}\} \leq k-1\Big) \geq \frac{k}{n+1}.$$

The event $\sum_{j=1}^n \mathbf{1}\{Y_j < Y_{n+1}\} \leq k-1$ is identical to $Y_{n+1} \leq Y_{(k)}$ (with $Y_{(n+1)} \coloneqq +\infty$). Thus $\mathbb{P}(Y_{n+1} \leq Y_{(k)}) \geq \frac{k}{n+1}$. For $k = k_\alpha = \lceil (n+1)(1-\alpha) \rceil$:

- If $k_\alpha \leq n$, $\widehat{B}_\alpha = \min(N, Y_{(k_\alpha)}) = Y_{(k_\alpha)}$, so $\mathbb{P}(Y_{n+1} \leq \widehat{B}_\alpha) \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geq 1-\alpha$.
- If $k_\alpha = n+1$, $\widehat{B}_\alpha = N$, and $Y_{n+1} \leq N$ holds with probability 1.

Combining with Theorem 4.4(ii) proves objective exactness with probability $\geq 1-\alpha$. □

A.5 Proof of Theorem 4.10 (Adaptive Split-Conformal Exactness)

Proof. Conditional on $\mathcal{D}_{\text{train}}$, $\hat{b}$ is fixed. The one-sided residuals $R_j = B_j^\dagger - \hat{b}(X_j)$ for $j \in \mathcal{D}_{\text{cal}} \cup \{\text{new}\}$ are exchangeable. Applying the rank argument from Theorem 4.9 gives $\mathbb{P}(R_{\text{new}} \le R_{(k_{\text{cal}})} \mid \mathcal{D}_{\text{train}}) \ge 1-\alpha$ for $k_{\text{cal}} \le n_{\text{cal}}$. The event $R_{\text{new}} \le q_\alpha$ implies $B_{\text{new}}^\dagger \le \hat{b}(X_{\text{new}}) + q_\alpha$. Since $B_{\text{new}}^\dagger \in \{B_{\min}, \dots, N\}$ is integer, this is equivalent to $B_{\text{new}}^\dagger \le \min(N, \max(B_{\min}, \lceil \hat{b}(X_{\text{new}}) + q_\alpha \rceil)) = B_\alpha(X_{\text{new}})$. For $k_{\text{cal}} = n_{\text{cal}} + 1$, safe fallback sets $B_\alpha = N \ge B_{\text{new}}^\dagger$ deterministically. Integrating over training data draws proves marginal exactness. □

A.6 Proof of Theorem 5.1 (Logarithmic Recovery Depth)

Proof. If $B_1 \ge B^\dagger(\theta)$, $\ell^*(\theta) = 1$, and $1 + [\lceil \log_r(B^\dagger/B_1) \rceil]_+ = 1 + 0 = 1$. If $B_1 < B^\dagger(\theta)$, let $\tilde{\ell} := 1 + \lceil \log_r(B^\dagger(\theta)/B_1) \rceil \ge 2$. Then $\tilde{\ell} - 1 \ge \log_r(B^\dagger/B_1) \implies B_1 r^{\tilde{\ell}-1} \ge B^\dagger(\theta)$. Since $B^\dagger$ is integer, $\lceil B_1 r^{\tilde{\ell}-1} \rceil \ge B^\dagger(\theta)$. Because $B^\dagger(\theta) \le N$, $B_{\tilde{\ell}} = \min(N, \lceil B_1 r^{\tilde{\ell}-1} \rceil) \ge \min(N, B^\dagger(\theta)) = B^\dagger(\theta)$. By minimality of $\ell^*(\theta)$, $\ell^*(\theta) \le \tilde{\ell}$. □

A.7 Proof of Theorem 5.2 (Cumulative Structural Work Bound)

Proof. Write $\ell^* := \ell^*(\theta)$. For $\ell^* = 1$, $\sum B_\ell = B_1 \le B^\dagger < \frac{r^2}{r-1} B^\dagger + 1$. For $\ell^* \ge 2$, minimality of ℓ^* implies $B_\ell < B^\dagger(\theta) \le N$ for all $\ell \le \ell^* - 1$. Hence the cap N never activates prior to stage ℓ^*, so $B_\ell = \lceil B_1 r^{\ell-1} \rceil \le B_1 r^{\ell-1} + 1$ for all $\ell \le \ell^* - 1$. At stage ℓ^*, $B_{\ell^*} \le \lceil B_1 r^{\ell^*-1} \rceil \le B_1 r^{\ell^*-1} + 1$ always holds. Summing across all stages: $\sum_{\ell=1}^{\ell^*} B_\ell \le B_1 \sum_{\ell=1}^{\ell^*} r^{\ell-1} + \ell^* = B_1 \frac{r^{\ell^*}-1}{r-1} + \ell^* < \frac{B_1 r^{\ell^*}}{r-1} + \ell^*$. Since stage $\ell^* - 1$ had no cap activation and $B_{\ell^*-1} < B^\dagger(\theta)$, we have $B_1 r^{\ell^*-2} \le \lceil B_1 r^{\ell^*-2} \rceil = B_{\ell^*-1} < B^\dagger(\theta)$. Multiplying by r^2: $B_1 r^{\ell^*} = r^2 (B_1 r^{\ell^*-2}) < r^2 B^\dagger(\theta)$. Substituting yields $\sum_{\ell=1}^{\ell^*} B_\ell < \frac{r^2 B^\dagger(\theta)}{r-1} + \ell^*$. Model size identities follow by affine summation. □

A.8 Proof of Theorem 5.4 (Two-Layer Guarantee)

Proof. Layer 1 is immediate from Theorem 4.10. Layer 2 is deterministic: Case A follows since $\text{LB}_{\text{full}}^{\text{relax}} \le J_\theta^* \le f_\theta(\hat{v}) = \text{UB} \implies 0 \le \text{UB} - J_\theta^* \le \text{UB} - \text{LB}_{\text{full}}^{\text{relax}} \le \varepsilon_{\text{cert}}$. Case B follows because at $B = N$, $P_\theta(\mathcal{N}) = P_\theta$. When the solver terminates with status `OPTIMAL`, ObjBound $\le J_\theta^* \le$ UB. For UB $\neq 0$, relative tolerance yields $\text{UB} - J_\theta^* \le \varepsilon_{\text{MIP}} |\text{UB}|$; for UB $= 0$, absolute tolerance yields $\text{UB} - J_\theta^* \le \varepsilon_{\text{abs}}$. Case C explicitly classifies uncertified timeouts. □

A.9 Proof of Theorem 5.7 (Perspective Lower-Bound Dominance)

Proof. For any feasible point in the perspective relaxation, $y_{h,i} \in [0,1]$. If $y_{h,i} \in (0,1]$, $t_{h,i} \ge x_{h,i}^2 / y_{h,i} \ge x_{h,i}^2$. If $y_{h,i} = 0$, semi-continuous bounds force $x_{h,i} = 0$, so $t_{h,i} \ge 0 = x_{h,i}^2$. Hence $t_{h,i} \ge x_{h,i}^2$ everywhere. Because $d_i \ge 0$, $\frac{\gamma}{2}(x_h^\top \boldsymbol{R} x_h + \sum d_i t_{h,i}) \ge \frac{\gamma}{2}(x_h^\top \boldsymbol{R} x_h + \sum d_i x_{h,i}^2) = \frac{\gamma}{2} x_h^\top \boldsymbol{\Sigma} x_h$. Minimizing over identical variable domains gives $\text{LB}^{\text{basic}} \le \text{LB}^{\text{persp}}$. Moreover, for binary $y \in \{0,1\}^N$, setting $t_i = x_i^2$ recovers the canonical problem exactly, so $\text{LB}^{\text{persp}} \le J^*$. Subtracting from UB yields $\text{UB} - \text{LB}^{\text{persp}} \le \text{UB} - \text{LB}^{\text{basic}}$, proving weak containment of certificate sets. □

B Experimental Reproducibility

All experiments are fully reproducible from the repository using parameters documented in the reproducibility manifest at `artifacts/final_reproducibility_manifest.json`. Hardware: AMD Ryzen 7 5800H (16 logical cores), 16 GB RAM, Windows 11. Solver: Gurobi 13.0.3 (`Threads`=16, `TimeLimit`=30 s, `MIPGap`=10^{-4} synthetic, 10^{-5} backtest). Canonical parameters: $\gamma = 5.0, \beta = 0.99, c_{\min} = 0.10, U_i = 0.40, L_i = 0.20, \tau^{\max} = 0.30, K = 3, F = \$10, a = 5\,\text{bps}, \delta = 0.02$.

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